

CAT: preserve the intended spaces

Ask for the behavior, not just a neat result

01 / THE SAS EXAMPLE

```
data test;
  length first middle last $8
         name1 name2 $24;
  input first $ middle $ last $;
  name1 = cat(first, middle, last);
  name2 = catx(' ', first, middle, last);
  datalines;
John Q Public
Jane . Doe
Bob . Smith
;
run;

proc print data=test noobs;
run;
```

ATTACH SOURCE OR PASTE IT AFTER THE PROMPT. START FRESH EACH TIME.

WHAT YOU WILL PRACTICE

Distinguish preserving source behavior from silently cleaning the data.

CAT keeps blanks in these character fields. CATX trims them and separates nonblank values. Both behaviors are intentional. Inspect the spaces as well as the readable names.

EXPECTED RESULT

Row	CAT: name1	CATX: name2
1	John____Q_____Public__	John Q Public
2	Jane_____Doe_____	Jane Doe
3	Bob_____Smith____	Bob Smith

Each underscore represents one stored blank in name1. name2 hides trailing storage padding. Input periods become blank middle fields. Both stored result columns have width 24.

Expected values checked in Python; SAS execution pending.

Prompt A: leave choices open

High degrees of freedom / Notice the decisions the model makes for you.

COPY THIS PROMPT + THE SAS EXAMPLE

Convert this SAS code to Python.

Show code in this response. Do not create or run files, or read existing Python files.

LOOK AT THE RESPONSE

Which library and method did it choose? Did it preserve the expected behavior? Compare the result with page 1 before accepting it.

TRY IT / THREE FRESH ATTEMPTS

Run A three times with the same SAS example and setup. Save each first response before asking follow-up questions. Record agreement as well as differences.

Bedrock A/B trials for this example are pending.

WHY CHOICES REMAIN OPEN

1 A familiar task

Joining name parts may prompt a familiar Python approach that prioritizes a readable name.

2 Two distinct behaviors

CAT preserves character blanks. CATX trims them and separates nonblank parts. Both results are intentional.

3 Preservation needs checking

A neat name cannot verify CAT. Make blanks visible before deciding whether the conversion matches.

ANTI-PATTERN TO AVOID

Anti-pattern: asking to 'clean up the names' during an equivalent conversion. That can invite space removal and change CAT. Request cleanup separately after checking the original behavior.

Prompt B: make the choices explicit

Low degrees of freedom / State the behavior, then ask for a check.

COPY THIS PROMPT + THE SAME SAS EXAMPLE

```
Convert this SAS CAT/CATX code to Python using pandas.
- Use first, middle, last in the original row order.
  Map the input periods to blank middle fields, not literal dots.
- Right-pad every input field to 8 characters with spaces.
- name1: concatenate the padded fields without trimming (CAT).
- name2: strip ordinary leading/trailing spaces from each field,
  skip blank fields, and join with one space (CATX).
- Store both result columns padded to width 24.
- Include assertions for both widths and all results below.
  For name1 checks, _ denotes one space:
  John__Q____Public__
  Jane____Doe____
  Bob____Smith__
  name2, with trailing padding hidden:
  John Q Public; Jane Doe; Bob Smith.
- Print name1 with spaces visible and name2 in readable form.
  Keep stored values unchanged; explain the difference briefly.

Show code in this response. Do not create or run files, or read
existing
Python files.
```

SAS REFERENCE

WHY THESE DETAILS HELP

1 Separate the operations

Give CAT and CATX their own rules so a general joining instruction does not hide the difference.

2 Define missing input

The middle fields are blank. Stating that prevents a dot or None from appearing in a name.

3 Make spaces countable

Widths and visible blank markers make stored spaces checkable. The readable display stays separate.

CHECK IT YOURSELF

Run the code. Compare each marked CAT string. Confirm both result columns have width 24. Check readable CATX names and verify a blank middle adds no extra separator.

Preserve the source behavior first. Make any requested cleanup a separate, explicit choice.